Data Validation & Cleaning

Sum of trip_distance	Sum of passenger_count	Sum of fare_amount	Sum of total_amount
3,55,90,704.05	1,19,76,598.00	18,16,21,025.14	27,25,94,981.54

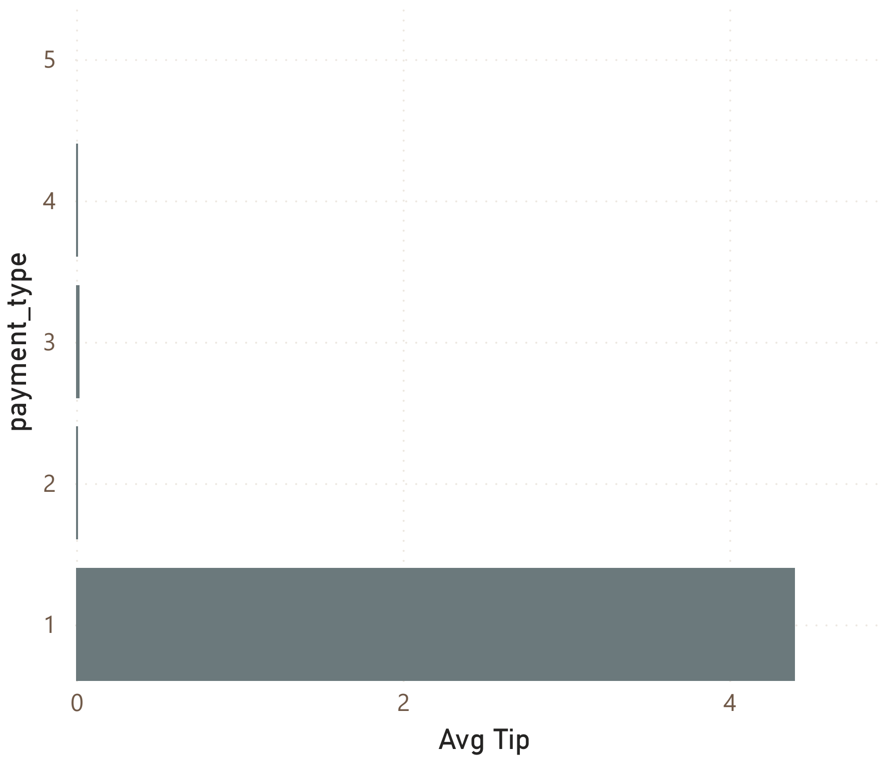
9M
Total Trips

29.75
Avg Fare

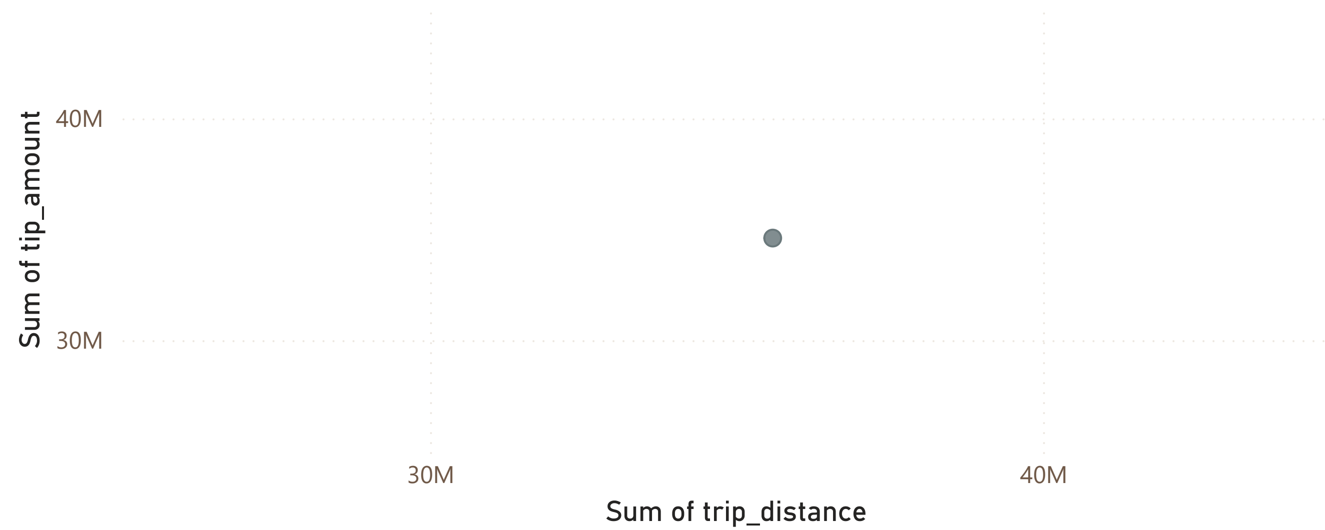
This dataset was cleaned by removing invalid trips where:
- Distance <= 0
- Fare <= 0
- Passenger count <= 0
- Dropoff time < Pickup time

This ensures analytical reliability.

Avg Tip by payment_type

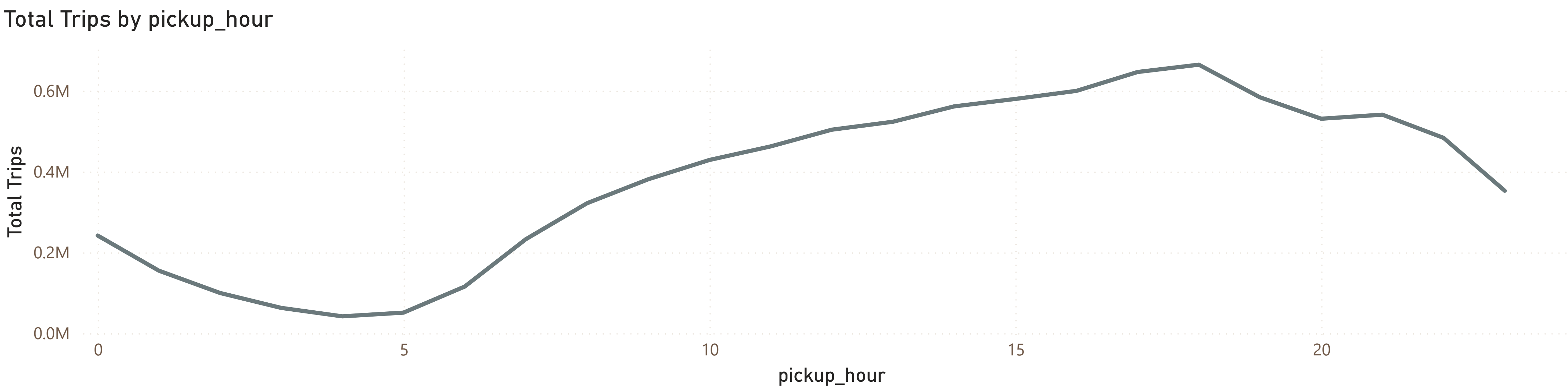


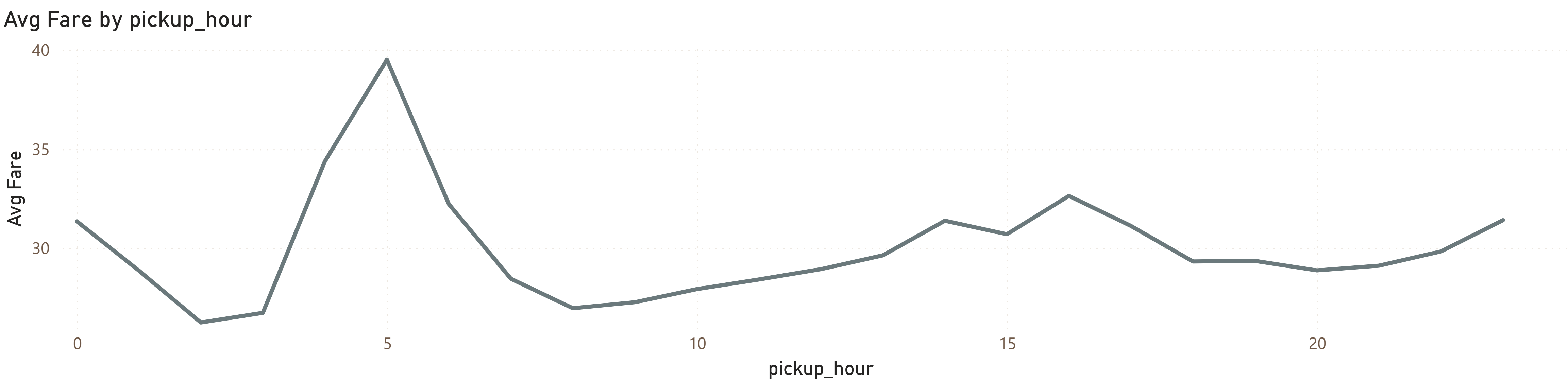
Sum of trip_distance and Sum of tip_amount



Hypothesis:
Longer trips produce higher tips.

Alternative:
Payment type influences tipping behavior more strongly.

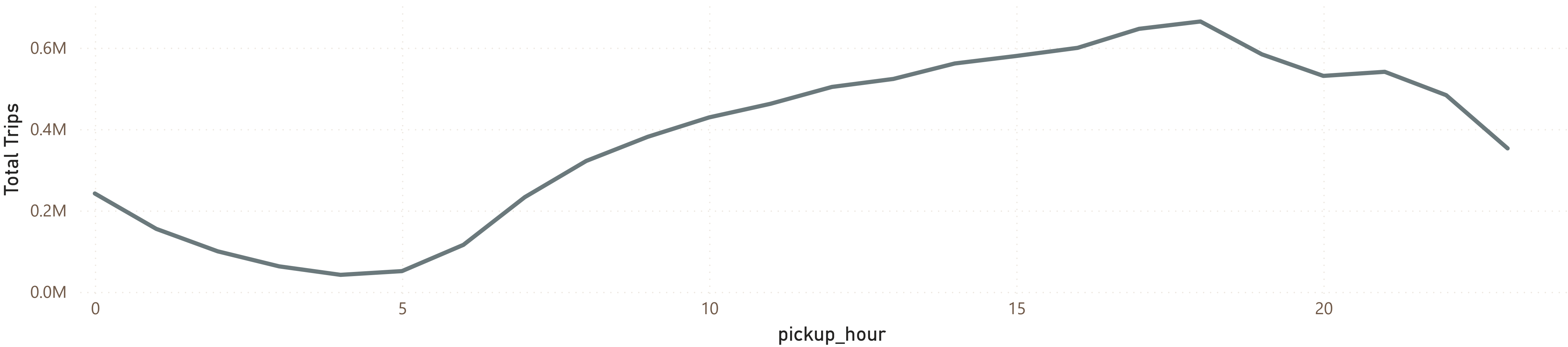




Patterns across hours reveal demand cycles.

Consistent peaks indicate structured behavior.

Total Trips by pickup_hour



9M

Total Trips

Peak-hour clustering indicates stress in system capacity.

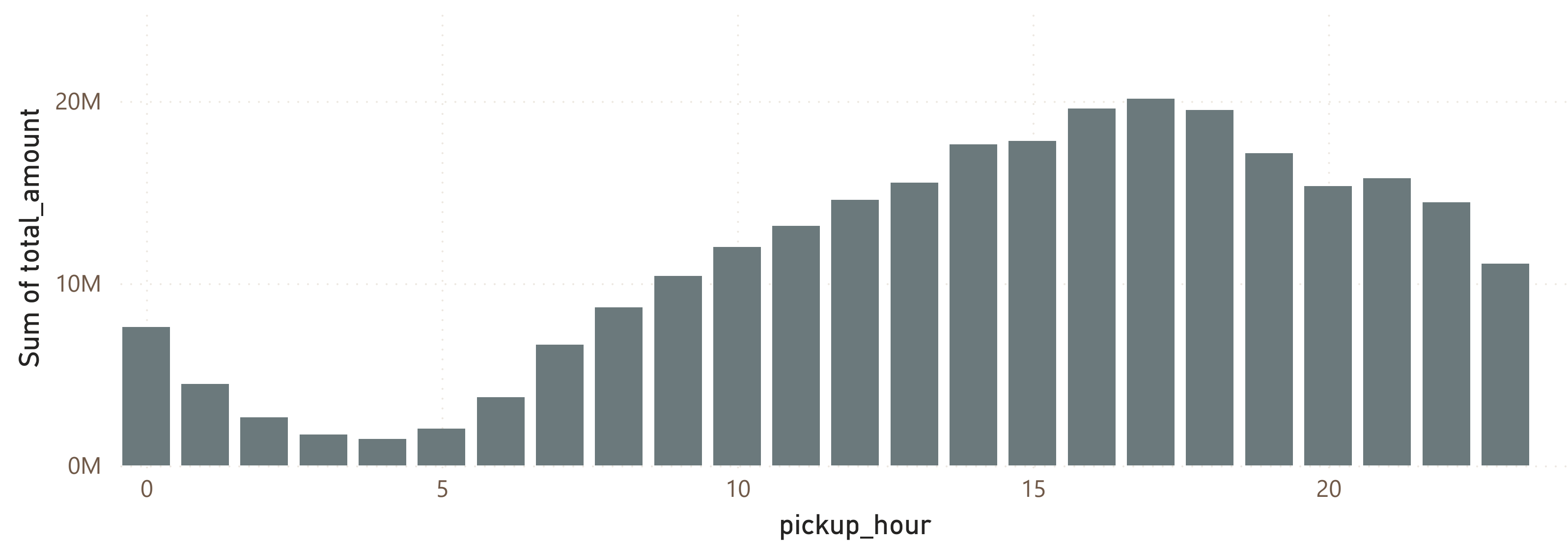
A 5% fare increase was simulated.

We compare system response before and after change.

272.59M
Actual Revenue

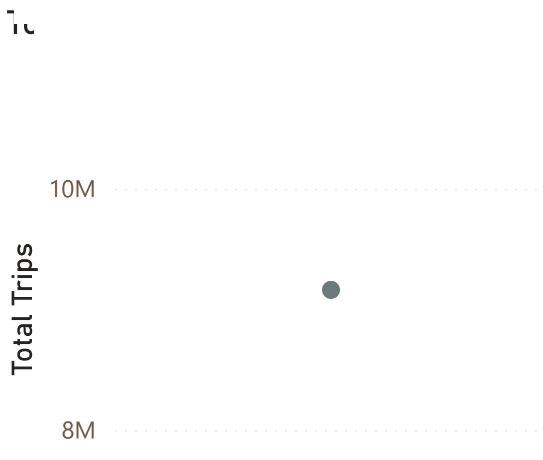
286.22M
Simulated Revenue

Sum of total_amount by pickup_hour

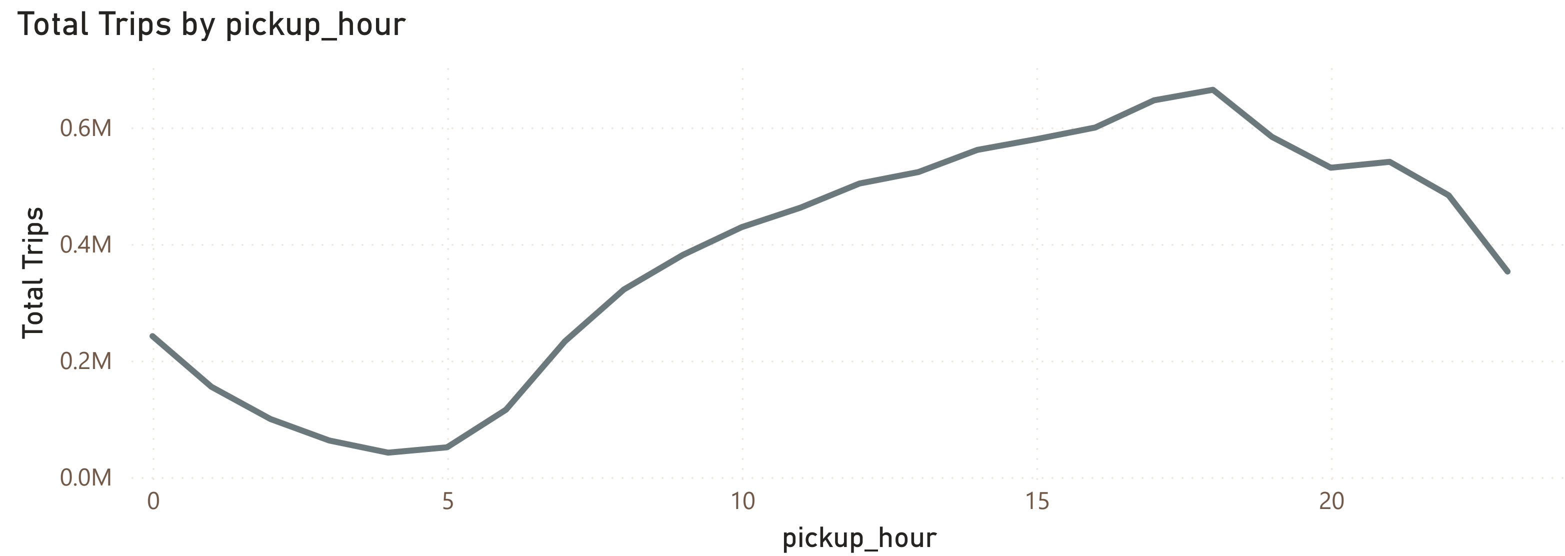


MISINTERPRETATION

wrong



correct



Total growth alone suggests stability.

However, concentration in peak hours reveals stress.

FINAL CONCLUSION (Act 7)

This analysis is based on cleaned and structured data.

The system shows cyclical demand with localized stress.

Simulation indicates moderate sensitivity to pricing changes.

Limitations:
- No external factors included
- Results depend on cleaning assumptions

Conclusion:
The system is stable overall but under pressure during peak periods.